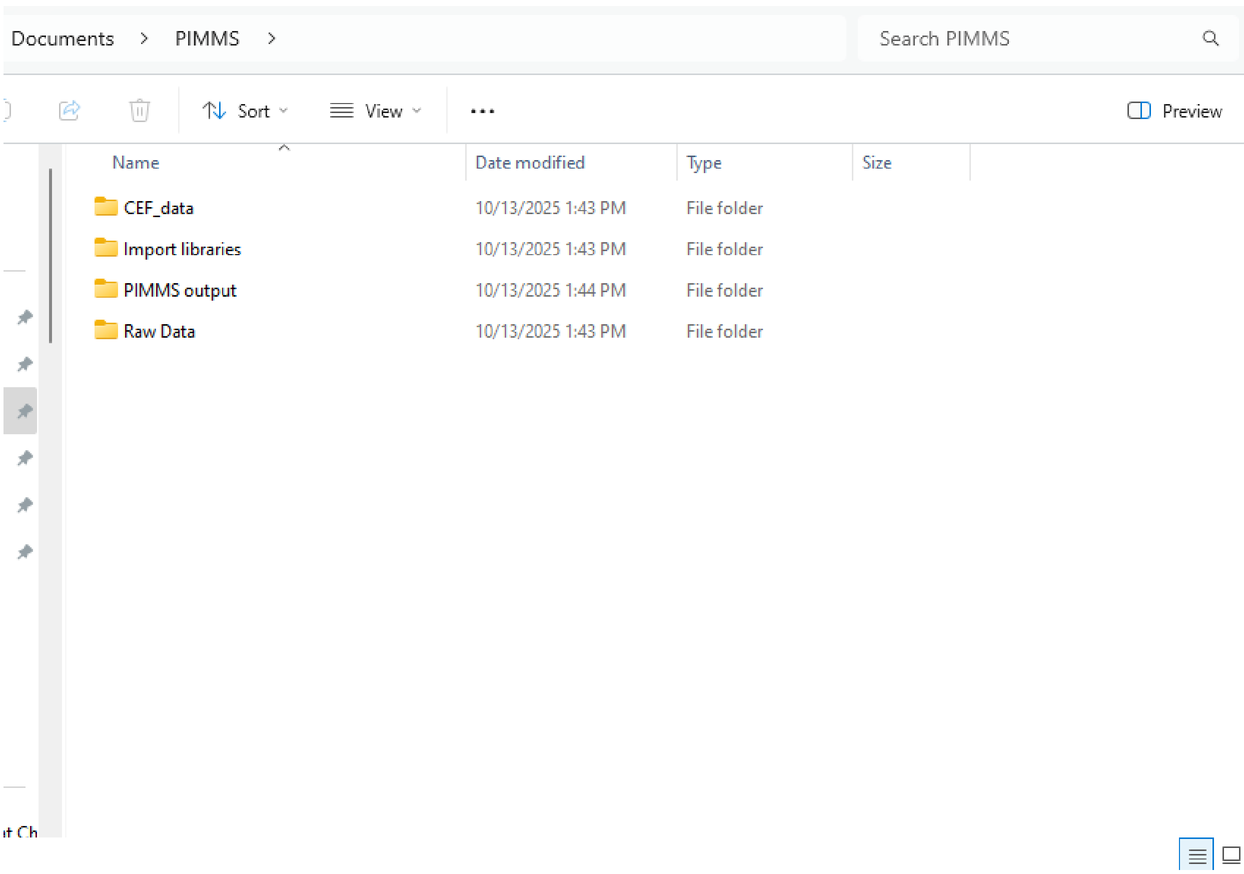
PIMMS Guide

Install

This is the page on which you are importing all the data that will be processed through PIMMS.

On install, PIMMS creates a user-friendly folder containing all data that will be needed for running in the Documents folder. Simply go to the Documents folder and look for "PIMMS".



Here, all libraries needed to score data are included. These libraries are included as default options although you are more than able to modify any library or use your own and simply specify their format.

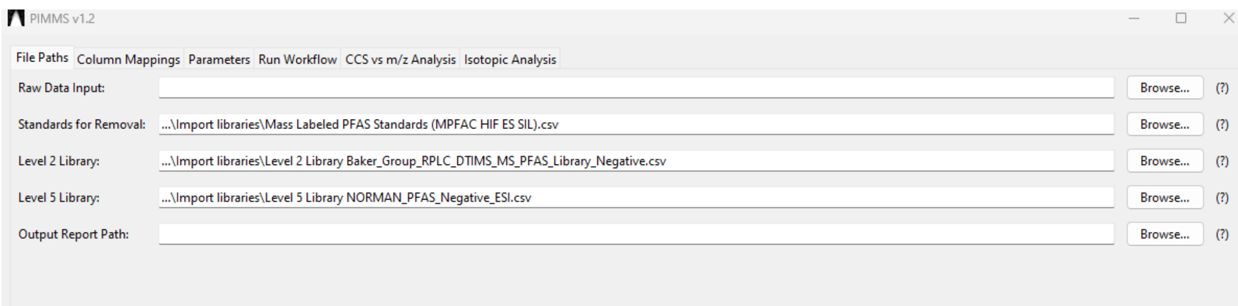
Tab: Files Path

The "Raw Data Input" is where you import your feature list. Your feature list must be a .csv file. This feature list must contain the following columns – ID, m/z, CCS, DT, RT, and then the intensities of each sample. Two sample groups should be present – an experimental and a control population.

"Standards for Removal". This corresponds to standards you wish to exclude from your final report; typically, mass labeled PFAS such as those contained in MPFAC-HIF-ES which PIMMS will remove by default. If you wish to remove other peaks, simply go to Documents>PIMMS>Import libraries>Mass Labelled… and edit or upload your own version. It simply needs a m/z column and a CCS column and any feature matching to that peak will be removed.

"Level 2 Library". This corresponds to previously ran standards for which a validated CCS and m/z exist. RT can also be used in scoring as well. By default, PIMMS is packaged with the Baker Lab's reversed phase liquid chromatography library consisting of 119 features representing 104 unique PFAS for which an analytical standard has been run. If you would like to add to the library then read in any library you would like, or add entries to the library.

"Level 5 Library". This corresponds to PFAS which exist in external compound libraries that possess a validated structure, however, no standard is available for further structural confirmation. By default, within PIMMS, a library of 573 features are scored against taken from the NORMAN Suspect List Exchange.[1] Note, NORMAN updates this database multiple times a year with the most recently discovered suspects. Prior to integration features are filtered by their preferred ionization mode, selecting only "RPLC_ESI-"and "ESI-".



"Output Report Path". This is the location your PIMMS report will be outputted. Hit browse and select any location and name you would like it to have.

Column Mappings

Here, we are mapping PIMMS to whatever your sample set is. This eliminates several sticking points when working with external data.

The "Metadata Column Letters (Raw Data)" section enables you to import data with any column ordering desired. By default, PIMMS loads with ID in A, RT in B, DT in C, CCS in D,

[1] Network, N. *et al.* (2025) *S0: Susdat: Merged norman suspect list: Susdat*, *Zenodo*. Available at: https://doi.org/10.5281/zenodo.15583645 (Accessed: 13 October 2025).

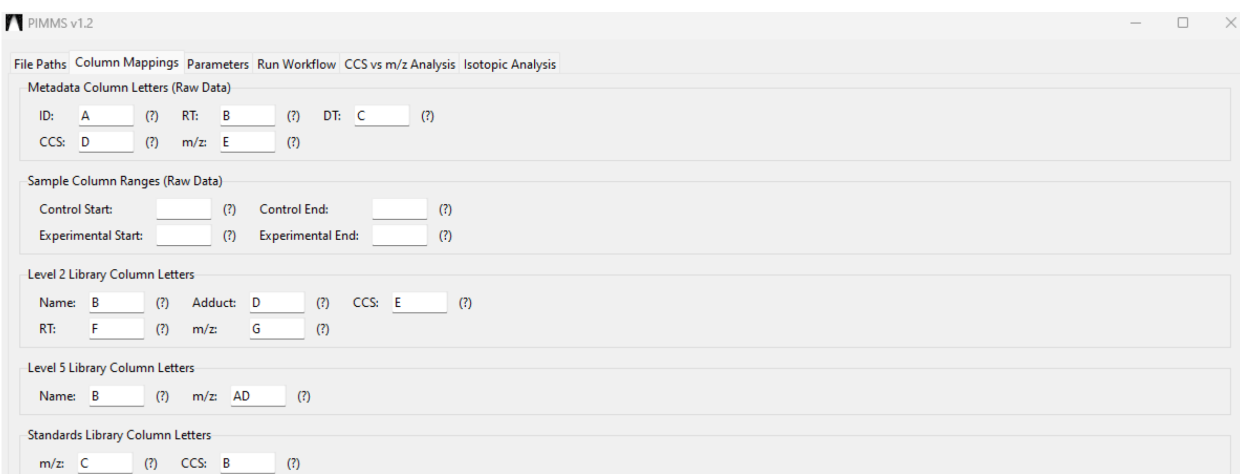
and m/z in E. However, this is entirely flexible, and if you have m/z in AZ, then just specify that.

Then; everyone's data is different. To overcome this, PIMMS allows you to specify the size of your data set. Simply specify the first column that your control population begins in, and then the last column. Then do the same for the experimental population.

"Level 2 Library Column Letters" corresponds to the mapping of the CCS library. PIMMS by default is configured to work with the Baker Lab's CCS library, however, if another library is used, simply modify the column mappings to align with your library. Exact column names are not important; PIMMS is configured to work with any column name so long as the correct column is specified.

"Level 5 Library Column Letters" corresponds to the mapping of the external library. PIMMS by default is configured to work with the NORMAN suspect library, however, if another library is used, simply modify the column mappings to align with your library. Again, exact column names are not important; PIMMS is configured to work with any column name so long as the correct column is specified.

"Standards Library Column Letters" corresponds to the mapping of the standards you wish to remove. PIMMS by default is configured to work with the file packaged in Documents>PIMMS>Import Libraries>Mass Labeled..., however, if another library is used, simply modify the column mappings to align with your library. Again, exact column names are not important; PIMMS is configured to work with any column name so long as the correct column is specified.



File Tab: Parameters

Here, we are specifying all the parameters necessary to score and filter data.

Tolerances and Filters

Mass Error (ppm) – the mass error that is considered a "match". For a run in which mass errors for known suspects are consistently within -10 - +10 ppm, I use an error of 15 ppm as peak alignment algorithms can shift the mass by a few ppm.

CCS Tolerance (%) – the error in CCS that is considered a "match" typically set to a max of 2%. Due to alignment algorithms, small errors can be propagated, so a <1% error in Browser is best targeted with a max 2% error.

RT Tolerance (min) – the acceptable bounds that a feature can have to be considered a match in scoring.

Minimum Intensity – the minimum intensity a sample's feature must have to be recorded. If the limit is 500 and there are 5 samples, Sample A: 50, Sample B: 50, Sample C: 5000, Sample D, 100, Sample E: 5000 the resulting matrix would be Sample A: 0, Sample B: 0, Sample C: 5000, Sample D, 0, Sample E: 5000.

Minimum RT, Maximum RT – sets the bounds for which a sample can be considered. By default PIMMS is set to 2 – 16 minutes as this corresponds to the "reliable" data range within the Baker Lab's RPLC method. However, you are welcome to set to any range desired.

Detection Frequency (%) – the minimum number of times a feature must be detected (after intensity filtering) to be counted. I.e. a detection frequency of 100% for a sample set of 5 samples, the feature must be detected in 5/5 samples above the intensity threshold.

Minimum m/z – maximum m/z – sets the mass range for analysis.

Mass Defect Filter – The bounds for which features are filtered by defect. A max upper bound of 0.09 and lower bound of -0.194 are set as this is found to align with the 5^{th} – 95^{th} percentile bounds for NORMAN's PFAS chemicals – including brominated, chlorinated, and iodine-containing features.

Blank Subtraction Method –

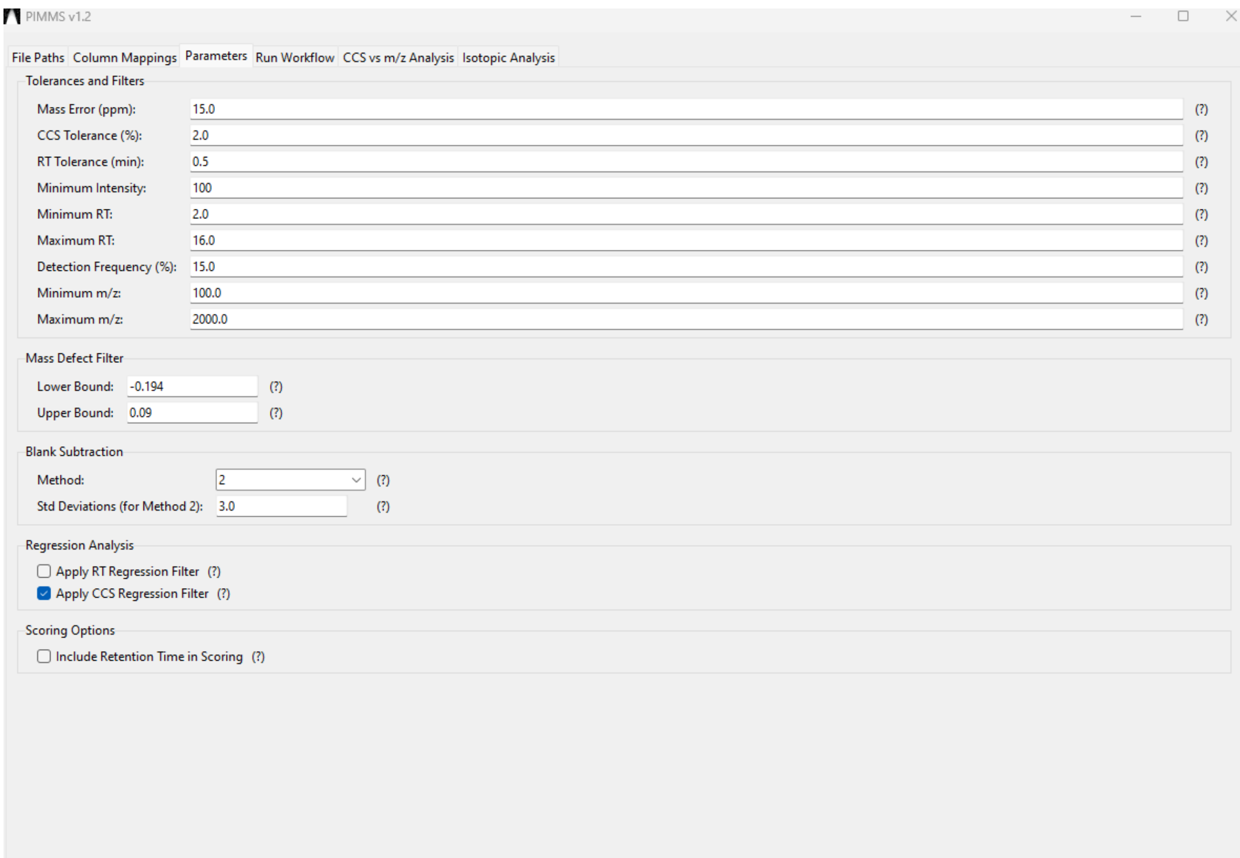
Method 2 – normal mean + standard deviations. Whatever the control range is, the mean and standard deviations will be calculated. Then depending on the value provided in the next box, a multiple of the standard deviations will be added. So Method 2 with 3 standard deviations would be $x - (\mu + 3\sigma)$

Method 1 – Whatever the most abundant feature is in the control population is subtracted from the experimental population. So, a control population of 10, 0, 5, 1000, would result in a value of 1000 which is subtracted from all experimental samples.

Regression Analysis – Here, the Level 2 library is used to create a 10th – 90th percentile logarithmic equation to determine where a typical PFAS sits in the CCS v m/z or RT v m/z relationship. This serves to greatly reduce the feature list, however, can eliminate PFAS in the feature list with unusually low degrees of fluorination or odd chromatography relative to their size.

Scoring Options

Include Retention Time in Scoring – This allows you to include retention time scoring in your analysis. This only applies to matching to the level 2 library. Typically this is left unchecked for larger more complex matrices due to the potential for retention time shifting with matrix effects.



Run Workflow

Hit Run PIMMS Workflow and it will parse your data and produce a PIMMS report to the output location specified on the File Paths location. Typically, a 100,000 feature report (10,000 features in 10 samples) will take around 1 minute to complete on a powerful computer. Running on a less powerful computer is possible, however, run times will increase.

CCS vs m/z Analysis

In my opinion, this is the most useful part of PIMMS – there's tons of features, but which ones are on a homologous series trend line with other features. To operate it, simply upload a PIMMS file to the "Load Experimental File" and specify the column ordering; by default, it is set to the PIMMS normal output.

Then, comparisons can be made to an external library to determine if a feature may align by a given repeat unit to another feature in a library. I.e. you see the 17-C PFCA, but the library only has 4-16 C PFCAs, here you would be able to see that your 17-C PFCA may lie in trend with the rest of the library if it is the same conformer.

Analysis Parameters:

Repeat Unit: set by default to CF2 but can be changed to any of 5 repeat units. If more are desired edit at PIMMS v1.2\modules\config.json.

PPM Error: max error range for a match in a repeat unit analysis. Again, for a 5 ppm run, set slightly wider to account for potential alignment algorithm shifting in feature processing algorithms.

Min. Series Points: The number of points that must be within a series for the series to be retained. Note a "point" is only considered valid if it possesses a point at least 10 Da away. I.e. a repeat unit of 50 and minimum series of 3 would meet the criteria of 100, 150, 200, but would fail if the points were 100, 100, 150, even though these features are in series with each other. However, 100, 150, 150, 200 would be reported as there are 3 points which have at least 10 Da separation. The minimum number of points is limited to 3 for obvious reasons.

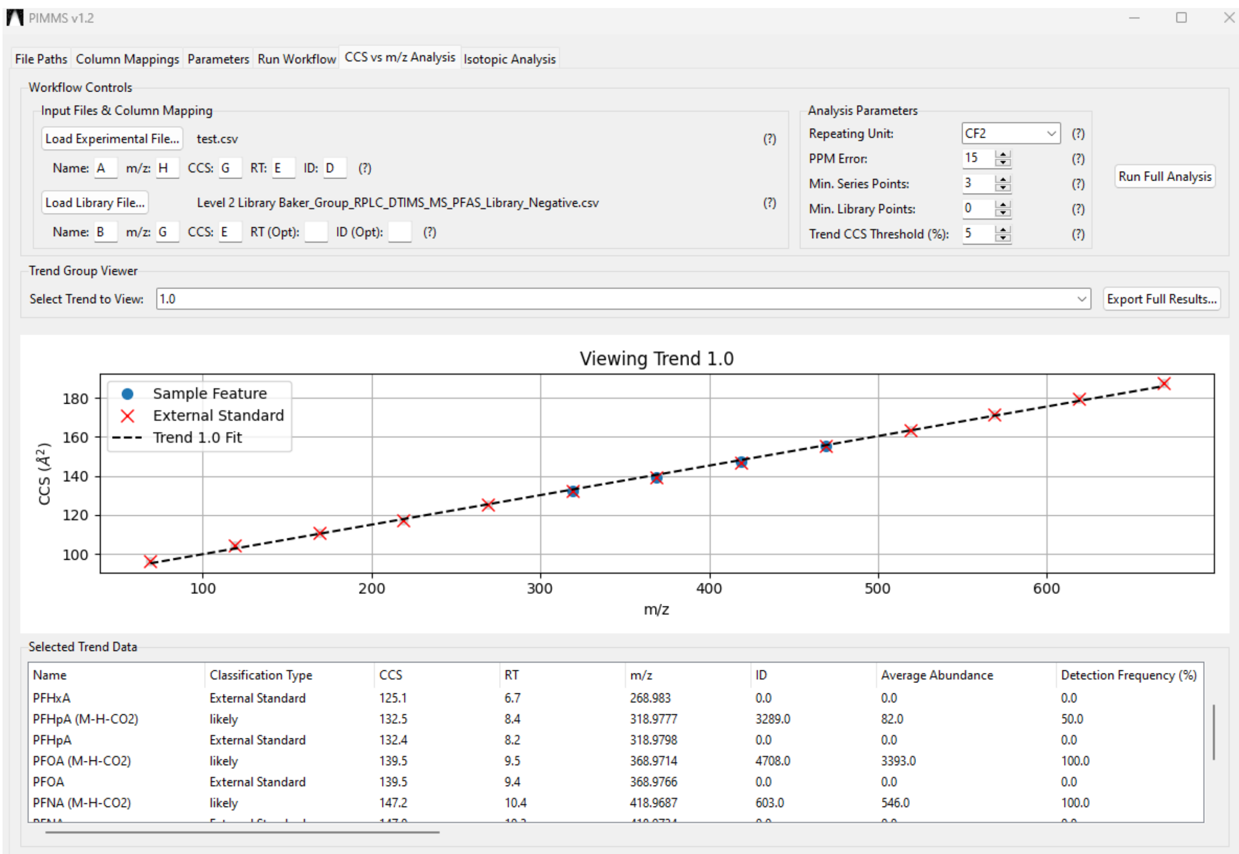
Min. Library Points: Gives you the option to only select trends that report a certain number of features originating from the library. Can be set at any value from 0 on up. A library point is represented as a red x in the plot

Trend CCS Threshold (%): This is the tolerance set for a point to be considered a "match" and included in the trend. By default, PIMMS sets this to 2% as this is found to be able to distinguish between branched and linear isomers while also retaining strong correlations. Note, PIMMS was developed using an Agilent 6560 DTIMS-QTOF platform with a IM resolving power of approximately 40-60, an instrument with a higher resolving power (i.e. a SLIM) may be able to push this value lower, although this would be achieved through trial and error in PIMMS.

Select Trend to View: This dropdown gives you the ability to view each CCS v m/z trend that is found within your data. Note, trends are separated first by their homologous series, and then if there is a subgroup of points. I.e. L-PFHxS, Br-PFHxS, Br-PFHpS, L-PFHpS, Br-PFOS, L-PFOS would be split into 1 group based on the CF2 repeat unit, then into 2 subgroups with the linear and branched isomers being displayed in separate trends provided their CCS are sufficiently separated from the linear isomers.

Export Full Results – This gives you the option to export your results as a .csv file. In the export all trends will be exported for convenience – PIMMS isn't the prettiest thing, if you wish to make a more aesthetically pleasing image of the new PFAS you find, this would be an easy way to do it.

Selected Trend Data – This shows all the trend points including points originating from the library. In addition, you can see the intensity in each sample for a given sample.



<u>Isotopic Analysis</u>

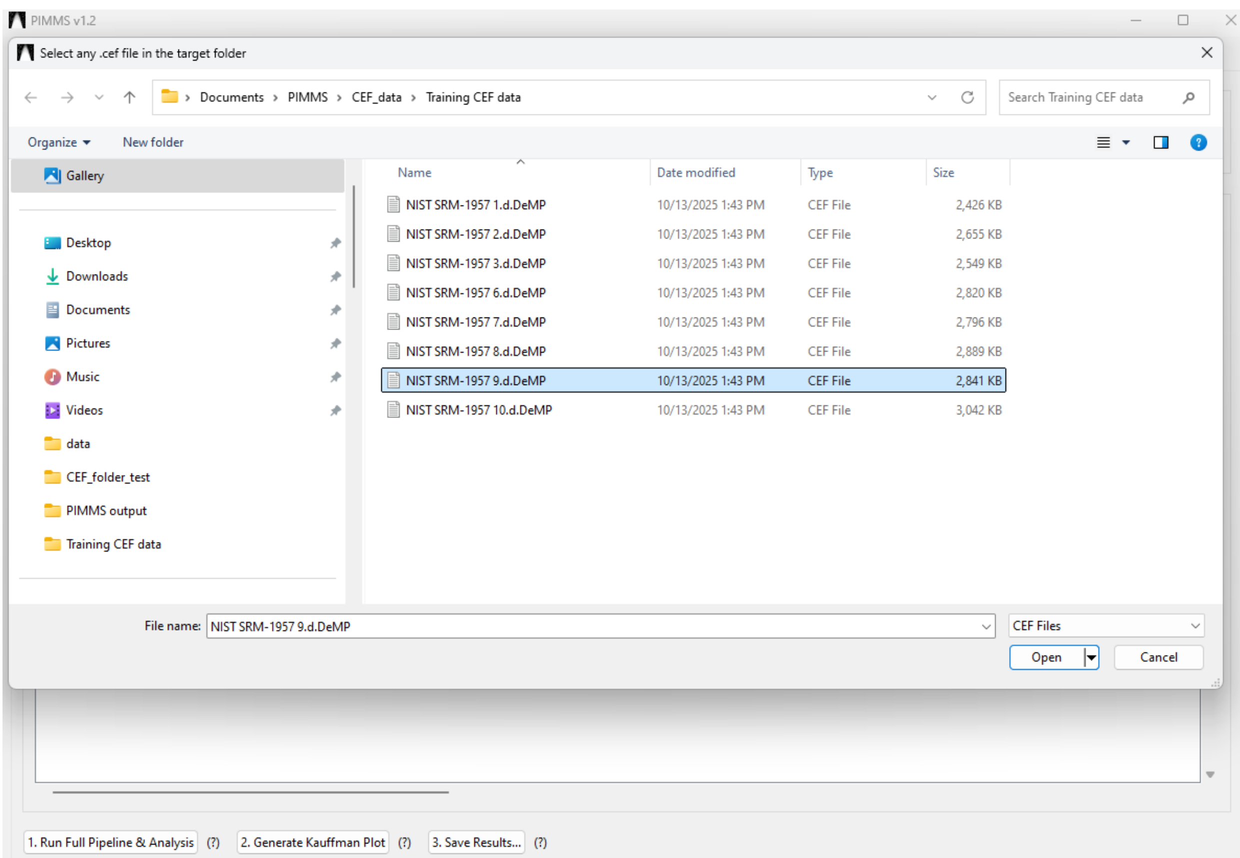
In PIMMS v1.2, this section is only possible for Agilent .d files.

This section allows you to perform isotopic analysis on features found in your PIMMS report to provide you with another level of information for structural identification. Note, only the most abundant peaks will be identified by Mass Profiler to have an isotopic packet. Mass

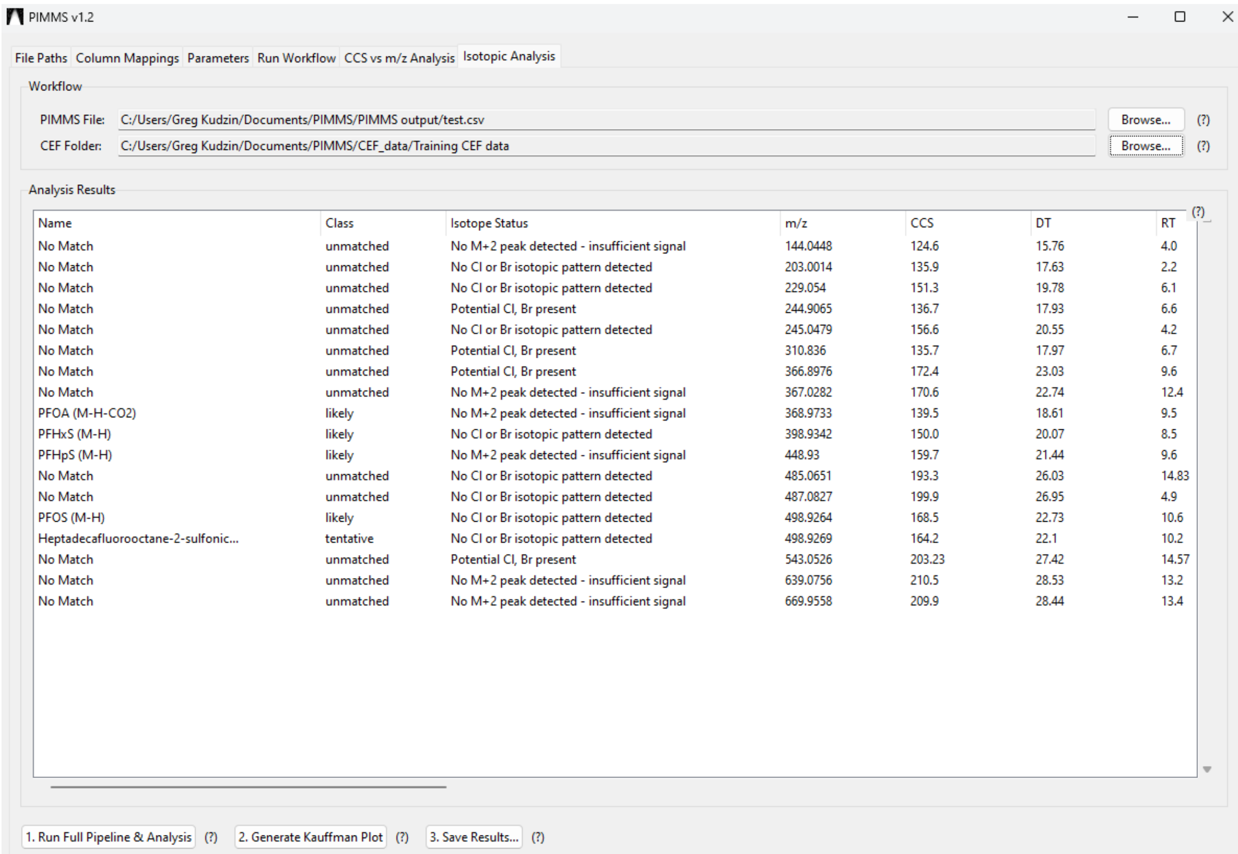
Profiler has a surprisingly high limit of detection within it's peak detection algorithm resulting in peaks being missed due to not being abundant enough. This applies to the M+1 peak of a feature – Mass Profiler may miss this unless the M+1 falls above the method limit of detection. For example, the PIMMS report for the training set of data has over 180 results, but only 18 features are detected to have both a M and M+1 peak. As such this presents a limiting factor.

To initiate analysis, first upload any PIMMS file you would like to analyze. Then, obtain .CEF files for the samples analyzed in that PIMMS report. To obtain .CEF files, go to Agilent's Mass Profiler, and upload your files. Once feature lists are processed using your desired method, then go to File>Export Each Sample to CEF>hit yes. A CEF file will be churned out for each sample uploaded and outputted to the location where you uploaded files to Profiler from, although this will be clear in the popup window in profiler.

CEF Folder: This is the location of the CEF files you would like to analyze. All you need to do is open the browse window and then locate the files you wish to process. PIMMS will read all the CEF files in the folder



Simply press "open" on any file in the folder and that will set the import directory as the folder in which the file is contained. Then PIMMS reads each CEF file in that folder and match to the column headers in the PIMMS report. I.e. it will match the “NIST SRM-1957 6.d.DeMP” CEF file with the column in the PIMMS report with the same name. This means you can have as many CEF files as you want, and PIMMS will pick those that match and process only those.



Name	Class	Isotope Status	m/z	CCS	DT	RT
No Match	unmatched	No M+2 peak detected - insufficient signal	144.0448	124.6	15.76	4.0
No Match	unmatched	No Cl or Br isotopic pattern detected	203.0014	135.9	17.63	2.2
No Match	unmatched	No Cl or Br isotopic pattern detected	229.054	151.3	19.78	6.1
No Match	unmatched	Potential Cl, Br present	244.9065	136.7	17.93	6.6
No Match	unmatched	No Cl or Br isotopic pattern detected	245.0479	156.6	20.55	4.2
No Match	unmatched	Potential Cl, Br present	310.836	135.7	17.97	6.7
No Match	unmatched	Potential Cl, Br present	366.8976	172.4	23.03	9.6
No Match	unmatched	No M+2 peak detected - insufficient signal	367.0282	170.6	22.74	12.4
PFOA (M-H-CO2)	likely	No M+2 peak detected - insufficient signal	368.9733	139.5	18.61	9.5
PFHxS (M-H)	likely	No Cl or Br isotopic pattern detected	398.9342	150.0	20.07	8.5
PFHpS (M-H)	likely	No M+2 peak detected - insufficient signal	448.93	159.7	21.44	9.6
No Match	unmatched	No Cl or Br isotopic pattern detected	485.0651	193.3	26.03	14.83
No Match	unmatched	No Cl or Br isotopic pattern detected	487.0827	199.9	26.95	4.9
PFOS (M-H)	likely	No Cl or Br isotopic pattern detected	498.9264	168.5	22.73	10.6
Heptadecafluorooctane-2-sulfonic...	tentative	No Cl or Br isotopic pattern detected	498.9269	164.2	22.1	10.2
No Match	unmatched	Potential Cl, Br present	543.0526	203.23	27.42	14.57
No Match	unmatched	No M+2 peak detected - insufficient signal	639.0756	210.5	28.53	13.2
No Match	unmatched	No M+2 peak detected - insufficient signal	669.9558	209.9	28.44	13.4

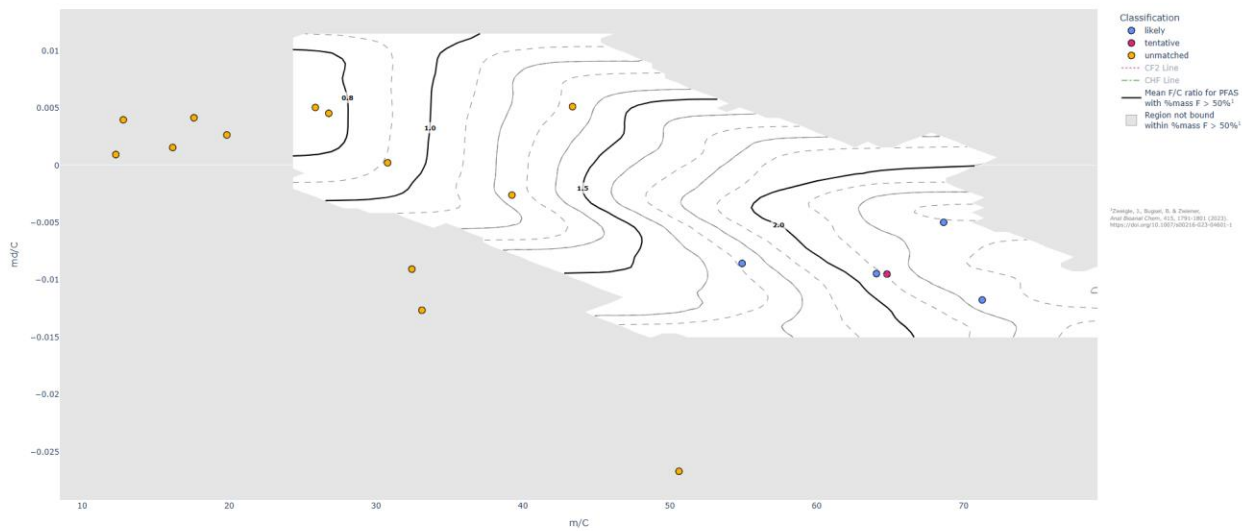
Isotope Status is a big addition to PIMMS. This column reports any M/M+2 ratio which is greater than 0.28, i.e. has the potential to be either chlorinated or brominated. The possible options here are “Potential Cl, Br present” (M/M+2 > 0.28), “No Cl or Br isotopic pattern present” (0<M/M+2< 0.28), “No M+2 peak detected – insufficient signal” (M/M+2 = 0).

Pred. F/C Ratio. This field enables you to get an estimate of the ratio of fluorines to carbons in the molecule, i.e. in the data PIMMS is distributed with, PFOS (C8F17SO3) has a predicted F/C ratio of 2.1 (true value 2.125).

Generate Kauffman Plot. Here you will create a Kauffman plot which gives the measure of mass defect with respect to isotopic distribution of the M to M+1 peak.

This plot has a topological style overlay which gives the F/C ratio for features. Note, this data is adapted from some incredible work done by Jonathan Zweigle at the University of

Tübingen.[2] In this paper PFAS with >50% of their mass composed by fluorines are modeled using theoretical distributions to get an analysis of how the ratio of fluorines to carbons changes with its position in a Kauffman plot. They found a R^2 of 0.88 for their ability to accurately predict F/C based on the theoretical model. As such it is incorporated into PIMMS to provide a useful tool for further structural elucidation.



Admittedly, Kaufmann plots may be new to even most non-target PFAS researchers. They were only formulated by Anton Kaufmann in 2022 as a method to prioritize PFAS from the surrounding biological matrix using only MS^1 data.[3] They prove to be an effective method to visualize PFAS and separate them as a result of fluorine's unique lack of an isotopic distribution resulting in heavily fluorinated substances having suppressed M+1 peaks relative to a typical hydrocarbon which possess far more carbons at the same mass and thus a more abundant M+1 peak due to the prevalence of ^{13}C.

To aid in the interpretation of the plot the CH_xF_{2-x} (CHF) line, and the CF_x (CF) line, are available to display, simply press them in the legend and they will display.

The contour lines demark the F/C ratio with solid lines at 0.8, 1, 1.5, 2 F/C. Grey lines are at every 0.1.

The grey space is the space that was not mapped by Zweigel *et. al.* due to the fact that no PFAS with a mass of fluorine >50% was found in this area. However, note that this does not cover all PFAS, only those with a heavy degree of fluorination.

Hovering over a point will display identifier information such as the point's CCS and what samples it appears in.

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[2] Zweigle J, Bugsel B, Zwiener C. *Anal Bioanal Chem.* 2023 doi: 10.1007/s00216-023-04601-1.

[3] Kaufmann, Anton et al. *Journal of AOAC International* 2022 doi:10.1093/jaoacint/qsac071

If a static image is desired of the plot, hover to the top right of the page and press the camera icon – a .png of the plot will be downloaded and dropped into your downloads folder.

If you wish to continue your analysis further, simply press the Save Results button. This will enable you to export the results as a .csv file to any location of your choosing. Within this report is contained the m/C value along with the md/C value to recreate your own Kaufmann plots. There is also the M/M+2 distribution value enabling a quick analysis of whether a peak may be chlorinated, brominated, multiply chlorinated and or brominated.